


Calculator continued :-

try mei sara sahi kaam likhte hai.

catch → mei error wali chize likhe.

```
try {  
    sahi code  
}
```

```
catch(e) {  
    error handling code  
}
```

Practice Code :-

```
let btns = document.querySelectorAll('button');  
let inpEl = document.querySelector('input');  
  
for(let btn of btns){  
    btn.addEventListener('click', function(e){  
        // let buttonText = e.target.innerText;  
        // inpEl.value = buttonText;  
  
        let buttonText = e.target.innerText;  
        inpEl.value = inpEl.value + buttonText;  
    })  
}
```

→ app.js

↳ this is not the final code but a practice code

final html code

index.html

```
<body>

  <div class="container">

    <input type="text">

    <div class="button-grid">

      <button class="grey-button">C</button>
      <button class="grey-button">**</button>
      <button class="grey-button">%</button>
      <button class="yellow-btn">/</button>

      <button>7</button>
      <button>8</button>
      <button>9</button>
      <button class="yellow-btn">*</button>

      <button>4</button>
      <button>5</button>
      <button>6</button>
      <button class="yellow-btn">-</button>

      <button>1</button>
      <button>2</button>
      <button>3</button>
      <button class="yellow-btn">+</button>

      <button class="zero">0</button>
      <!-- <button>0</button> -->
      <button>.</button>
      <button class="yellow-btn">=</button>
    </div>
  </div>

  <script src="app.js"></script>
</body>
```

in my code, i applied loop on all buttons, so when i run js code, the loop should automatically be applied to all button.

Yes, you're correct! When the JavaScript code runs, the loop automatically applies the `click` event listener to all buttons. Let's break it down step-by-step to clarify why this happens:

```
let btns = document.querySelectorAll('button');
for (let btn of btns) {
  btn.addEventListener('click', function (e) {
    let buttonText = e.target.innerText;
    inpEl.value = inpEl.value + buttonText;
  });
}
```

[Copy code](#)

Key Steps:

1. `document.querySelectorAll('button')`:
 - This selects all **button** elements on the page and returns a `NodeList` (similar to an array) containing each button.
2. `for (let btn of btns)`:
 - The `for...of` loop iterates over every button in the `NodeList`.
 - For every button (`btn`), the loop runs once and executes the body of the loop.
3. `btn.addEventListener('click', function (e) { ... })`:
 - Inside the loop, an event listener is added to each button.
 - When the button is clicked, the `click` event triggers the `function(e)` that is defined here.

By the end of the loop, every button has been assigned its own `click` event listener.

Why Doesn't the Loop Wait for Button Clicks?

The loop is **not waiting for button clicks**. Instead:

1. The loop completes execution immediately when the script runs, iterating through all buttons.
2. The `addEventListener` function registers the `click` event for each button. This registration is done in advance, so when a user clicks a button later, the event listener is ready to respond.

What Happens on Page Load?

When the page loads:

1. The script runs.
2. All buttons are processed by the loop, and each button is registered with its own `click` event listener.

For example, if you have 16 buttons in the HTML, the loop runs 16 times and assigns the same `click` event listener function to each button.

What Happens When You Click a Button?

1. The **assigned event listener** for that specific button runs.
2. Inside the event listener, the following happens:
 - `e.target.innerText` retrieves the text of the clicked button.
 - `inpEl.value` is updated by appending the button's text to the current input value.

